Advaith Renjith

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EDUCATION

Rutgers University–New Brunswick, School of Engineering | B.S. Mechanical Engineering **Class of 2030**

HONORS

Sir Isaac Newton Physics Exam: 9th internationally of 3,800+ (2025) | **USAPhO:** Distinction, top 400 of 6,000 (2025)
NASA App Development Challenge: National Winner, 200+ teams (2025) | **MIT Beaver Works:** 4th of 100 teams (2025)
NJ State Science Fair: 2nd Place, Computer Science, flood model (2026); Air Force Award, drone-swarm RL (2025)

PUBLICATION

Beyond the Haystack: Sensitivity to Context in Legal Reference Recall. Third author. NLLP Workshop at EMNLP 2025.

- Evaluation framework for long-context LLM recall on three naturalistic legal-reference tasks built from U.S. court-opinion briefings; exposed behavior missed by synthetic benchmarks such as Needle in a Haystack.

RESEARCH & ENGINEERING PROJECTS

Rutgers Formula Racing: Chassis Optimizer Chassis subteam **2026**

- Wrote a ground-structure optimizer that picks frame tubes and sizes for torsional stiffness per pound, using a Timoshenko beam surrogate (adjoint sensitivities, Sherman–Morrison–Woodbury updates, tabu and integer search) checked against the SolidWorks beam study.
- The optimized frame scored about 3,100 lb·ft/deg at 75.5 lb in the team's SolidWorks torsion procedure, roughly 44% more stiffness per pound than the reference frame; now working on the 2028 frame in Ansys.

BLUEBIRD Survey Drone Designed and simulated; prototype next **2026**

- Designed a 1.8 m fixed-wing drone for conservation and agriculture in FreeCAD, with a printable center body and removable foam wings; produced full CAD and an 86-page engineering report.
- Developed Android RGB/thermal perception and mission logic with independent ArduPilot control and failsafes; evaluated the design with SU2 CFD, CalculiX FEA and ArduPilot software-in-the-loop tests.

Active Aerodynamics RC Testbed Independent; current design analysed, not yet built **2023–2026**

- Designed a 1/10 BT46B-style fan car: twin blowers on a skirted plenum (predicted 18.4 N, 0.87 of weight), a DRS wing, and servo-actuated ride height, all generated from one parameter file within a $500 budget.
- Checked it with scripted FreeCAD CAD and motion sweeps, SU2 CFD, CalculiX FEA and a controls software-in-the-loop simulation; earlier 1/8-scale prototype ran its ECU on a repurposed smartphone.

Physics-Informed Flood Forecasting Model Science fair research **2025–2026**

- Built a PyTorch graph transformer for flood forecasting with the 1D Saint-Venant continuity equation in its loss; in project benchmarks against a finite-difference reference, reached 117× faster inference with 97% mass-conservation adherence.

Cable-Driven Soft Robotic Arm Independent; built and bench-tested **2024–2026**

- Built and bench-tested a three-cable TPU continuum arm for 3D bending; added camera-based hand-pose control and a trajectory visualizer; developing a VLM + heuristic solver controller.

Furball: Tactical Flight Sim Browser game, live on CrazyGames **2026**

- Built and shipped a Three.js air-combat game with 100 aircraft across five eras, four mission modes, progression and private multiplayer rooms; 636 players in its first two days of launch.

Kestrel Mini R4 Bicopter and **SA-2 Samson VTOL** CAD prototypes; Kestrel build in progress **2026**

- Designed a FreeCAD bicopter with twin lift motors, tilt servos, 40 printed parts and a 42-route wire schedule (est. 665 g, 1.82 thrust-to-weight); modeled a 1:15 tilt-nacelle VTOL with coaxial rotors and a staged test plan.

Axiom: Offline Math Notebook macOS desktop application **2026**

- Built an offline notebook with typeset equations, 2D/3D graphs and a reference-PDF pane, plus a 2D/3D truss solver that checks determinacy, solves member forces and grades answers.

NASA App Development Challenge, Team Luna Vista National Winner **2025**

- Modeled the Artemis II flight path in Unreal Engine 5 using orbital mechanics and C++ shaders; optimized antenna scheduling for the visualization.

SKILLS & TRAINING

CAD & simulation	FreeCAD, SolidWorks; SU2 and OpenFOAM (CFD), CalculiX (FEA), ArduPilot SITL
Hardware	3D-printed TPU and structural parts, cable actuation, RC electronics and wiring layouts
Software	Python, C++, JavaScript/Three.js, Unreal Engine 5, physics-informed ML, reinforcement learning
Safety	OSHA-10